Mathematical Appendix: Advanced Recycling Macroeconomic Model

This document provides the formal derivation of the theoretical model presented in the main thesis. The model evaluates the economic viability of advanced recycling technologies under environmental constraints.

1. Endogenous Variables and Parameters

- Y_t: Final Output.
- R_t, $V_{1,t}$, $V_{2,t}$: Virgin resource, Standard recycled input, Advanced recycled input.
- M_t: CES Material Aggregate.
- Z_t, E_t: Pollution stock and current emissions.
- $u_t \in [0,1]$: Mitigation/Abatement level.

2. Production and Environmental Damage The representative firm produces output subject to a persistent environmental damage that reduces productivity.

$$Y_t = Ae^{-Z_t} M_t$$

Emissions are generated during production and can be reduced by choosing a mitigation level u_t:

$$E_t = (1 - u_t) d_E Y_t$$

The pollution stock accumulates over time, with a natural decay rate δ_Z:

$$Z_{t+1} = (1 - \delta_Z) Z_t + E_t$$

3. The Material Input Aggregator (CES) Inputs are imperfect substitutes. Advanced recycling ($V_{2,t}$) has a higher technological efficiency ($A_2 > A_1$). The aggregator is defined as:

$$M_t = [wR_t^\rho + \gamma(A_1 V_{1,t})^\rho + (1 - w - \gamma)(A_2 V_{2,t})^\rho]^{\frac{1}{\rho}}$$

with $\rho = \frac{\sigma-1}{\sigma}$.

4. Firm's Profit Maximization The firm maximizes current profit by choosing the optimal mix of inputs and the mitigation level, facing convex costs for materials and technical costs for abatement:

$$H_t = \Lambda(u_t)Y_t - a_R R_t^2 - bV_{1,t}^2 - cV_{2,t}^2$$

Where $\Lambda(u_t)$ represents the net marginal value of output, accounting for mitigation costs and residual emission taxes:

$$\Lambda(u_t) = 1 - du_t^{\bar{d}} - t(1-u_t)d_E$$

padroneggi la catena delle derivate senza saltare passaggi.

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5. First Order Conditions (FOCs)

Preparation: Derivate of Y_t and M_t

First, we derive H_t with respect to Y_t:

$$\frac{\partial H_t}{\partial Y_t} = \Lambda(u_t) = 1 - du_t^{\bar{d}} - t(1-u_t)d_E$$

Let us define the inner CES term as S_t:

$$S_t \equiv wR_t^\rho + \gamma(A_1V_{1,t})^\rho + (1-w-\gamma)(A_2V_{2,t})^\rho, \quad M_t = S_t^{1/\rho}$$

The general derivative of M_t with respect to any input $x_t \in \{R_t, V_{1,t}, V_{2,t}\}$ is:

$$\frac{\partial M_t}{\partial x_t} = \frac{\partial}{\partial x_t}(S_t^{1/\rho}) = \frac{1}{\rho}S_t^{\frac{1}{\rho}-1}\frac{\partial S_t}{\partial x_t}$$

Since:

$$S_t^{\frac{1}{\rho}-1} = S_t^{\frac{1-\rho}{\rho}} = (S_t^{1/\rho})^{1-\rho} = M_t^{1-\rho}$$

We get:

$$\frac{\partial M_t}{\partial x_t} = \frac{1}{\rho} M_t^{1-\rho} \frac{\partial S_t}{\partial x_t}$$

Because $Y_t = Ae^{-Z_t} M_t$, we have:

$$\frac{\partial Y_t}{\partial x_t} = Ae^{-Z_t} \frac{\partial M_t}{\partial x_t}$$

FOC for Virgin Resource (R_t)

Step 1. Chain Rule on H_t:

$$\frac{\partial H_t}{\partial R_t} = \Lambda(u_t) \frac{\partial Y_t}{\partial R_t} - 2a_R R_t = 0$$

Calculating the intermediate derivatives:

$$\frac{\partial Y_t}{\partial R_t} = Ae^{-Z_t} \frac{\partial M_t}{\partial R_t}$$

$$\frac{\partial M_t}{\partial R_t} = \frac{1}{\rho} M_t^{1-\rho} \frac{\partial S_t}{\partial R_t}$$

$$\frac{\partial S_t}{\partial R_t} = w\rho R_t^{\rho-1}$$

Substituting:

$$\frac{\partial M_t}{\partial R_t} = \frac{1}{\rho} M_t^{1-\rho} (w\rho R_t^{\rho-1}) = w M_t^{1-\rho} R_t^{\rho-1}$$

Therefore:

$$\frac{\partial Y_t}{\partial R_t} = Ae^{-Z_t} w M_t^{1-\rho} R_t^{\rho-1}$$

Final FOC for R_t:

$$\Lambda(u_t) Ae^{-Z_t} w M_t^{1-\rho} R_t^{\rho-1} = 2a_R R_t$$

FOC for Standard Recycling ($V_{1,t}$)

Step 1. Chain Rule on H_t:

$$\frac{\partial H_t}{\partial V_{1,t}} = \Lambda(u_t)\frac{\partial Y_t}{\partial V_{1,t}} - 2bV_{1,t} = 0$$

Calculating the intermediate derivatives:

$$\frac{\partial Y_t}{\partial V_{1,t}} = Ae^{-Z_t}\frac{\partial M_t}{\partial V_{1,t}}$$

$$\frac{\partial M_t}{\partial V_{1,t}} = \frac{1}{\rho}M_t^{1-\rho}\frac{\partial S_t}{\partial V_{1,t}}$$

$$\frac{\partial S_t}{\partial V_{1,t}} = \gamma\rho(A_1V_{1,t})^{\rho-1}A_1$$

Substituting:

$$\frac{\partial M_t}{\partial V_{1,t}} = \frac{1}{\rho}M_t^{1-\rho}(\gamma\rho A_1(A_1V_{1,t})^{\rho-1}) = \gamma M_t^{1-\rho}A_1(A_1V_{1,t})^{\rho-1}$$

Therefore:

$$\frac{\partial Y_t}{\partial V_{1,t}} = Ae^{-Z_t}\gamma M_t^{1-\rho}A_1(A_1V_{1,t})^{\rho-1}$$

Final FOC for $V_{1,t}$:

$$\Lambda(u_t)Ae^{-Z_t}\gamma M_t^{1-\rho}A_1(A_1V_{1,t})^{\rho-1} = 2bV_{1,t}$$

FOC for Advanced Recycling ($V_{2,t}$)

Step 1. Chain Rule on H_t:

$$\frac{\partial H_t}{\partial V_{2,t}} = \Lambda(u_t)\frac{\partial Y_t}{\partial V_{2,t}} - 2cV_{2,t} = 0$$

Calculating the intermediate derivatives:

$$\frac{\partial Y_t}{\partial V_{2,t}} = Ae^{-Z_t}\frac{\partial M_t}{\partial V_{2,t}}$$

$$\frac{\partial M_t}{\partial V_{2,t}} = \frac{1}{\rho}M_t^{1-\rho}\frac{\partial S_t}{\partial V_{2,t}}$$

$$\frac{\partial S_t}{\partial V_{2,t}} = (1 - w - \gamma)\rho(A_2 V_{2,t})^{\rho-1} A_2$$

Substituting:

$$\frac{\partial M_t}{\partial V_{2,t}} = \frac{1}{\rho} M_t^{1-\rho}((1 - w - \gamma)\rho A_2 (A_2 V_{2,t})^{\rho-1}) = (1 - w - \gamma) M_t^{1-\rho} A_2 (A_2 V_{2,t})^{\rho-1}$$

Therefore:

$$\frac{\partial Y_t}{\partial V_{2,t}} = Ae^{-Z_t}(1 - w - \gamma) M_t^{1-\rho} A_2 (A_2 V_{2,t})^{\rho-1}$$

Final FOC for $V_{2,t}$:

$$\Lambda(u_t) Ae^{-Z_t}(1 - w - \gamma) M_t^{1-\rho} A_2 (A_2 V_{2,t})^{\rho-1} = 2cV_{2,t}$$

FOC for Mitigation Level (u_t)

Assuming Y_t does not depend directly on u_t, we take the derivative of H_t with respect to u_t:

$$\frac{\partial H_t}{\partial u_t} = Y_t \frac{d\Lambda(u_t)}{du_t} = 0$$

Since $\Lambda(u_t) = 1 - du_t^{\bar{d}} - t(1 - u_t)d_E$, deriving term by term gives:

$$\frac{d\Lambda}{du_t} = -d\bar{d}u_t^{\bar{d}-1} + td_E$$

Therefore:

$$\frac{\partial H_t}{\partial u_t} = Y_t(-d\bar{d}u_t^{\bar{d}-1} + td_E)$$

Final FOC for u_t:

$$Y_t(-d\bar{d}u_t^{\bar{d}-1} + td_E) = 0$$

With $Y_t > 0$:

$$d\bar{d}u_t^{\bar{d}-1} = td_E$$

If $\bar{d} > 1$, the internal solution is:

$$u_t^* = \left(\frac{td_E}{d\bar{d}} \right)^{\frac{1}{\bar{d}-1}}$$

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6. Steady State In the long run, the pollution stock stabilizes when emissions equal the natural absorption capacity:

$$Z^* = \frac{E^*}{\delta_Z}$$

$$Z^* = \frac{(1 - u^*)d_E Y^*}{\delta_Z}$$